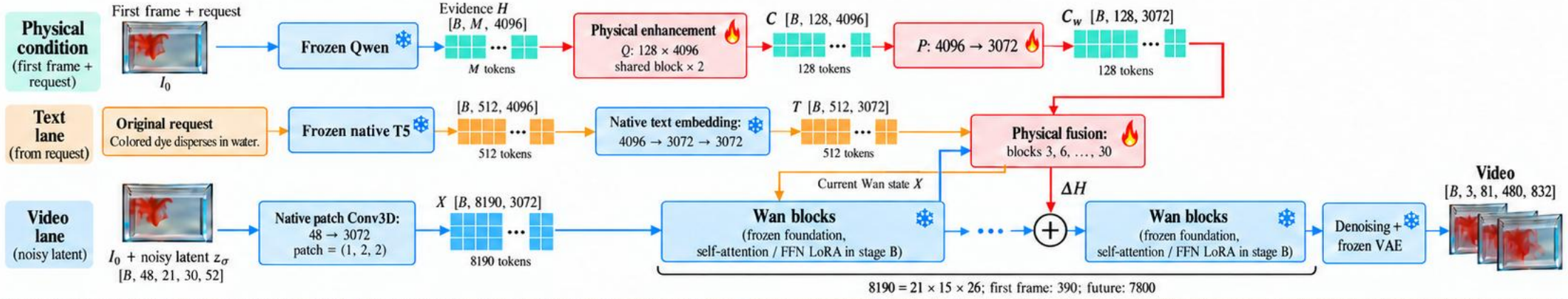
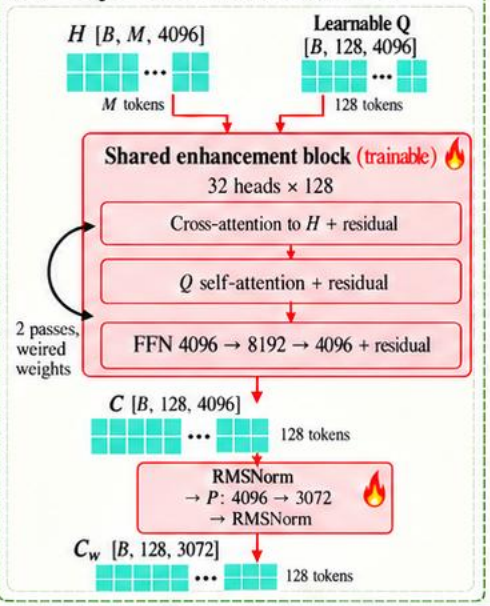
Physical Write Distillation V6

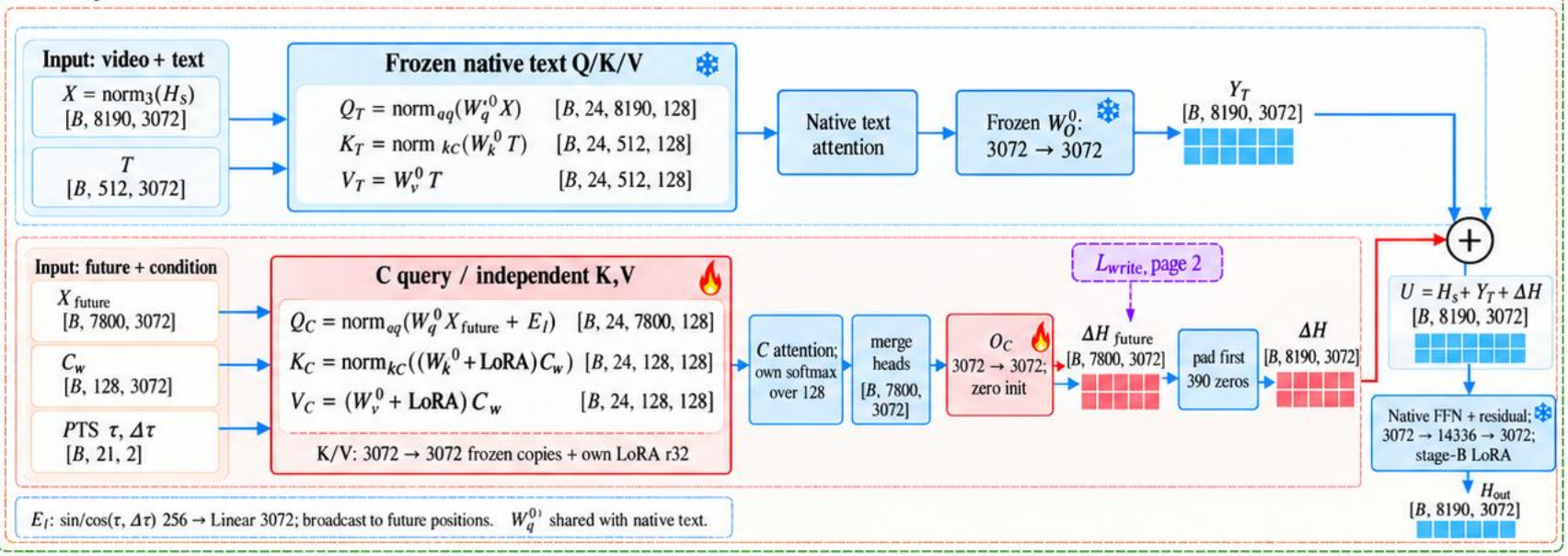
Physical enhancement + physical fusion | Proposed architecture



(a) Physical enhancement



(b) Physical fusion at one Wan block



Stage A: train enhancement + P + C K/V LoRA + O_C + time; freeze Wan → **Stage B:** continue A + warm up Wan self-attention and FFN LoRA

Frozen foundation (pretrained) | Trainable new parameters | Native Wan module | Evidence / C (physical)
Native text module | Physical branch (trainable) | Training only (loss, etc.)

B : batch; M : Qwen context tokens; τ : video presentation time; σ : diffusion noise. Proposed design, not a trained result.

Physical Write Distillation V6 — Training

Real-video weak targets -> actual physical write | Two objectives

(a) Offline target construction — training data only

1. Video-conditioned physical relations (where/when, from real clip)

Real training clip: 81 RGB frames



28 observed frames $[B, 28, 3, 480, 832]$

Caption / candidate WISA phys_law

"A red dye disperses in water due to diffusion."

Frozen Qwen

Relation sentences $R \le 4$

- r_1: dye diffuses
- r_2: spreads in water
- r_3: forms plume
- r_4: mixes with water

Relation-to-video attention

$N_v = T_q H_q W_q$ (actual Qwen grid)

N_v video tokens

$\alpha_{relation}$ $[B, R, N_v]$

$\alpha_{background}$ $[B, 1, N_v]$

Positive difference + normalization

Soft evidence g $[B, R, N_v]$

Crop/resize + PTS resampling (no learned router)

g_{grid} $[B, R, 21, 15, 26]$

$21 \times 15 \times 26 = 8190$

Future g $[B, 7800, R]$

7800 tokens

Merge to layer-l teacher target

$D_l^* = g \otimes e_l$

D_l^* $[B, 7800, 3072]$

Cached target; stop gradient

2. Relation text semantics (what physics, from same sentences)

Same relation sentences $(R \le 4)$

- r_1: dye diffuses
- r_2: spreads in water
- r_3: forms plume
- r_4: mixes with water

Frozen native T5

T_r $[B, R, L_r, 4096]$

L_r text tokens

Native Wan text embedding $4096 \to 3072 \to 3072$

Linear $4096 \to 3072$

Linear $3072 \to 3072$

$[B, R, L_r, 3072]$

Mean over valid text tokens

Mean (masked)

t_r $[B, R, 3072]$

Frozen layer-l native W_v^0 and W_o^0 (weights only)

W_v^0, W_o^0

e_l $[B, R, 3072]$

$[B, 7800, R] \times [B, R, 3072]$

g: where/when; e_l: physical conditional semantics

(b) Full-video flow matching

Real latent z $[B, 48, 21, 30, 52]$

$\varepsilon \sim N(0, I)$

$\sigma \sim U(0, 1)$

$z_\sigma = (1-\sigma)z + \sigma\varepsilon$ (first frame fixed)

Wan + physical C (same model as page 1)

Predicted v_θ $[B, 48, 21, 30, 52]$

Target $\varepsilon - z$ $[B, 48, 21, 30, 52]$

$$\mathcal{L}_{FM} = \text{mean}_{\text{valid_future}} \|v_\theta - (\varepsilon - z)\|^2$$

Valid future positions: $20 \times 15 \times 26 = 7800$ patch tokens

v_θ is denoising velocity, not physical motion speed.

(c) Direct supervision of the ACTUAL write

C_w $[B, 128, 3072]$

Current Wan X $[B, 8190, 3072]$

Physical fusion: C K/V + attention + O_C

Actual delta H_l $[B, 7800, 3072]$

Actual residual addition before FFN

pad 390 zeros

H_s + native text output $[B, 8190, 3072]$

Native FFN + residual: stage-B LoRA

Cached D_l^* $[B, 7800, 3072]$

stopgrad

$$s_l = \max(\text{RMS}(D_l^*), 10^{-3})$$

$$\mathcal{L}_{\text{write}} = \text{mean SmoothL1}\left(\frac{\text{delta } H_l - \text{stopgrad}(D_l^*)}{s_l}\right)$$

Target-only scale; future tokens; average over 10 injection layers.

$$\mathcal{L} = \mathcal{L}_{FM} + \lambda_{write}(1-\sigma)\mathcal{L}_{write}$$

Initial $\lambda_{\text{write}} = 0.02$

backprop (train only)

- C enhancement (parameters)
- P (parameters)
- C K/V LoRA ($r = 32$)
- O_C (parameters)
- Wan self-attention LoRA (stage B)
- Wan FFN LoRA (stage B)

Two-stage training (new modules and LoRA only)

A : train C enhancement, P, C K/V LoRA, O_C, time projection; Wan frozen

B : continue A + Wan self-attention / FFN LoRA ($r = 32$)

Official pretrained foundations; fresh project modules and LoRA.

B = batch; $R \le 4$ relation sentences; L_r = text length; N_v = teacher video tokens
l = injection layer; D^* = training target; delta H = actual residual write.

Teacher branch is offline training-only: inference uses first frame + original request.
Proceed architecture; token maps are weak annotations.

Frozen (pretrained) · Evidence / token · Text / sentence · Physical computation · Training only · Training loss (only)